`unitTransformDist`: A Distribution-Agnostic Toolkit for Unit-Interval Models via the Odds-Inverting Transform

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Abstract

Data restricted to the open interval $(0, 1)$ – proportions, rates, probabilities – are commonly modelled in R with `betareg` (beta regression), `simplexreg` (simplex regression), or the recently released `gkwdist` (generalized Kumaraswamy family). Each of these packages targets one specific parametric family. This paper introduces `unitTransformDist`, which instead constructs a $(0, 1)$-valued distribution from an arbitrary base distribution on $(0, \infty)$ via the transform $Y = 1/(1 + X)$, taking the base distribution's density, CDF, quantile, and random-generation functions directly as arguments rather than selecting from a fixed registry. This design lets users plug in R's built-in distributions or their own custom research distributions – including ones with no closed-form quantile function – without modifying the package. Beyond this generality, we document and empirically correct a subtle but consequential implementation error common to profile-likelihood confidence intervals: freezing nuisance parameters at their unconstrained maximum-likelihood estimate while profiling a parameter of interest, rather than re-optimizing them at each candidate value. A reproducible Monte Carlo study included in the package ($n = 60$, 150 replications, a two-parameter gamma base distribution) shows this mistake reduces actual 95% coverage to approximately 59%, and that the corrected procedure restores coverage to approximately 94%. The package also provides Monte Carlo utilities for studying estimator bias and several bias-correction methods (jackknife, bootstrap, Firth-type). We describe the package's design, its relationship to existing CRAN alternatives, and its current limitations.

1 Introduction

Continuous data confined to $(0, 1)$ arise across many applied fields: disease prevalence, vote shares, budget allocations, and normalized indices are all commonly analyzed as proportions. The dominant approach in R is beta regression via `betareg` [Cribari-Neto and Zeileis, 2010], which models the mean and precision of a beta-distributed response. Alternatives include `simplexreg` for the simplex distribution [Zhang et al., 2016] and the recent `gkwdist` package, which implements the seven-member generalized Kumaraswamy family with C++-accelerated likelihood and derivatives [Carrasco et al., 2010]. A related but architecturally different idea appears in `Newdistns`, which takes an arbitrary baseline CDF G and layers a fixed menu of generator families on top of it (Marshall–Olkin G, exponentiated G, beta G, and related constructions).

`unitTransformDist` addresses a narrower transform – the odds-inverting map $Y = 1/(1 + X)$, the inverse of the classical odds transform – but removes the restriction to a fixed set of base distributions entirely: the base distribution's d/p/q/r functions are supplied directly by the caller. We do not claim this transform itself is new; our contribution is (a) a general-purpose implementation that works unmodified with any base distribution the user can write in R, including one with no closed-form quantile function, and (b) a documented and empirically verified fix to

a profile-likelihood implementation mistake that we believe is easy to make and likely present, in some form, in other unit-interval fitting code.

2 Design considerations

The package targets applied statisticians who already have a candidate base distribution in mind – possibly a distribution not available in any existing package – and want to fit its unit-transform version without first registering it anywhere. Three design decisions follow from this.

First, `utd()` accepts `ddist`, `pdist`, `qdist`, and `rdist` as plain R functions, matching the base-R d/p/q/r convention. `qdist` and `rdist` are optional: if omitted, quantiles are obtained by numerically inverting `pdist` with `stats::uniroot()`, and random draws are obtained by inverse-CDF sampling through that same numerical quantile function. This lets a user supply only a density and a CDF for an entirely custom distribution and still get a fully functional object.

Second, every exported, user-facing function validates its inputs at the boundary (domain checks, named-vector checks, `NA` checks) and raises an informative error via `stop()`, rather than propagating `NaN` silently. Internal helpers (prefixed with a dot) skip this validation, since their callers have already validated it.

Third, the package uses base R's S3 object system and implements the generics a user already expects: `print()`, `summary()`, `coef()`, `logLik()`, `vcov()`, `AIC()`, `BIC()`. `summary()` methods return an object of a `summary.*` class rather than printing directly, and a corresponding `print.summary.*` method does the printing – matching the convention used by `summary.lm()` in base R, and avoiding `cat()`/`print()` calls inside computational functions.

3 Typical workflow

```
library(unitTransformDist)
d <- utd(dgamma, pgamma, qgamma, rgamma, params = list(shape = 2.5))
y <- rutd(500, d)
fit <- fit_utd_mle(y, dgamma, start = c(shape = 1), lower = c(shape = 1e-6))
fit
profile_likelihood_ci(fit, "shape")
```

A fully custom, unregistered distribution follows the identical pattern:

```
dcustom <- function(x, rate, log = FALSE) dexp(x, rate = rate, log = log)
pcustom <- function(q, rate, lower.tail = TRUE) pexp(q, rate, lower.tail = lower.tail)
d2 <- utd(dcustom, pcustom, params = list(rate = 1.5))
qutd(0.5, d2) # numerically inverted, no qdist supplied
```

4 The profile-likelihood correction: technical details and empirical evidence

For a model with parameter vector $\theta = (\theta_1, \ldots, \theta_p)$, the profile likelihood for θ_1 at a candidate value t is defined as

$$L_{\text{prof}}(t) \;=\; \max_{\theta_2,\ldots,\theta_p} L(t, \theta_2, \ldots, \theta_p), \tag{1}$$

i.e., the nuisance parameters must be re-optimized, not fixed, at every candidate t. An implementation that instead evaluates the likelihood at $(t, \hat{\theta}_2, \ldots, \hat{\theta}_p)$ – the nuisance parameters frozen at their

unconstrained MLE – computes something else entirely, and the resulting "confidence interval" from inverting a χ^2 threshold against this quantity has no guaranteed coverage properties.

`unitTransformDist::profile_likelihood_ci()` performs the correct re-optimization at every grid point via nested calls to `stats::optim()`. We also provide `safe_profile_likelihood()`, a version of the same routine that is deliberately factored out to accept any user-supplied negative log-likelihood function, independent of the rest of the package, so the fix can be audited or reused on its own.

To quantify the practical impact, `compare_profile_likelihood_coverage()` runs both versions – naive fixed-nuisance and corrected – on the same simulated datasets and reports empirical 95% coverage:

```
true_dist <- utd(dgamma, pgamma, qgamma, rgamma,
                 params = list(shape = 2.5, rate = 1))
compare_profile_likelihood_coverage(
  true_dist, c(shape = 2.5, rate = 1), dgamma,
  start = c(shape = 1, rate = 1), param_name = "shape",
  n = 60, n_sim = 150, lower = c(shape = 1e-6, rate = 1e-6), seed = 1)
```

Method	Coverage (95% target)	n_{valid}
Naive (fixed nuisance)	0.587	150
Corrected (re-optimized)	0.940	150

Table 1: Empirical coverage of the naive versus corrected profile likelihood, $n = 60$, 150 Monte Carlo replications, gamma base distribution with shape = 2.5, rate = 1.

The naive version's actual coverage (58.7%) falls far short of its nominal 95% target; the corrected version (94.0%) is close to nominal. This is, to our knowledge, the first published empirical quantification of this specific mistake's coverage impact for a unit-interval model, and we present it as the package's central contribution. We flag as experimental (and explicitly do not recommend for production use) an additional analytic bias-correction routine, `bias_correct_analytic()`, whose derivation exploits that the transform's Jacobian is parameter-free, so the classical Cox and Snell O(1/n) correction [Cox and Snell, 1968] transfers unchanged from the base distribution's likelihood; the current finite-difference implementation of the required third derivatives has not yet been validated against a closed-form case and should be treated as a documented but unverified idea rather than a working tool. We report this limitation explicitly rather than omitting it.

5 Performance and limitations

The package has no compiled code and depends only on `stats` and `graphics`; every operation is pure R. This keeps the dependency footprint minimal but means large-sample maximum likelihood fitting and the Monte Carlo/bootstrap utilities scale linearly in the number of `optim()` calls, which can be slow for very large `n_sim` or `n_boot`; we have not benchmarked or parallelized this. Where quantiles or random draws require numerical inversion (`qdist`/`rdist` omitted), each call to `qutd()`/`rutd()` performs one `uniroot()` search per element, which is markedly slower than a closed-form quantile function. The package currently provides no regression, mixture, or censored-data extensions of the transform; these are natural next steps but are not part of this release, and the current version's mathematical verification effort has focused specifically on the profile-likelihood correction rather than on every function in the package.

6 Discussion and conclusion

`unitTransformDist` does not claim to be the first package to generalize across base distributions (`Newdistns` already does this for a different family of transforms), nor does it claim the odds-inverting transform itself is new. Its value is narrower and, we think, still useful: a minimal-dependency, any-distribution implementation of this specific transform, paired with a concrete, reproducible demonstration that a common profile-likelihood implementation error meaningfully damages confidence interval coverage on this model class. We see the distribution-agnostic `safe_profile_likeliho` routine as reusable beyond this package's own scope, and we hope documenting this fix explicitly – with the coverage numbers to show it matters – is useful to other package authors implementing profile likelihood for their own models.

References

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